EE 152-BASIC ELECTRONICS

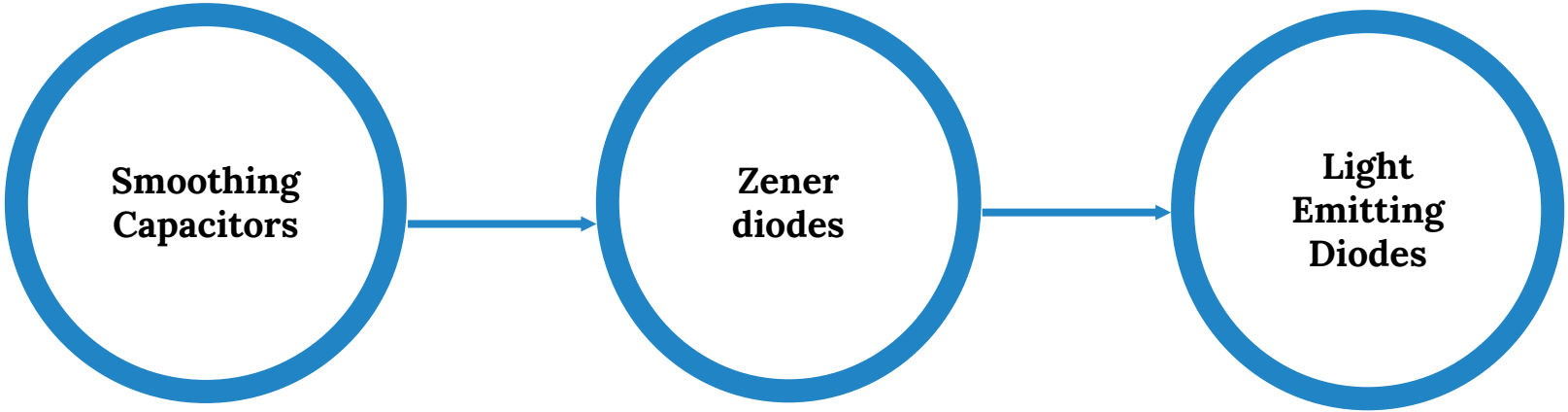
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CHAPTER FOUR

APPLICATIONS OF SEMI-CONDUCTOR DIODES - 2

What to expect?





1. SMOOTHING CAPACITOR

A smoothing capacitor is used to filter out the AC component present in a rectified DC voltage.



Smoothing capacitor

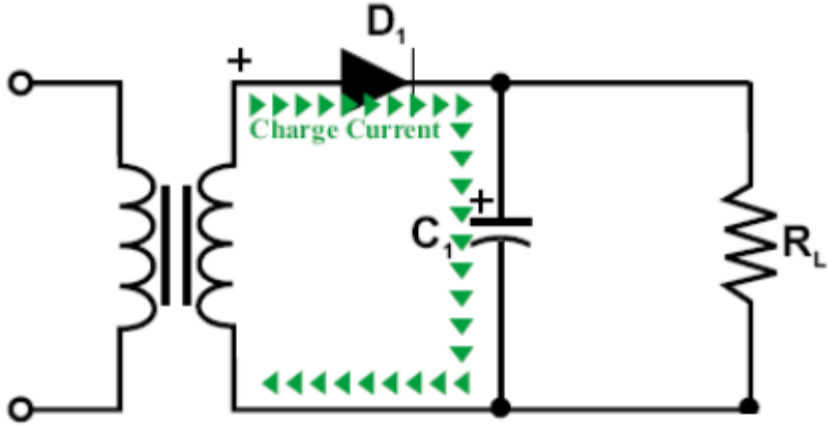
- The output obtained by the rectifier is not pure dc but it contains some ac components along with the dc output.
- These ac components are called as ripples, which are undesirable or unwanted.
- To minimize the ripples in the rectifier output, filter circuits are used.
- These circuits are normally connected between the rectifier and load.
- The smoothing capacitor is a filter that can be used for filtering the ac component

1. Half-Wave Rectifier with Capacitor Filter

- **A capacitor filter is connected in parallel with the load. The rectifier circuit is supplied from a transformer.**

Positive Half Cycle

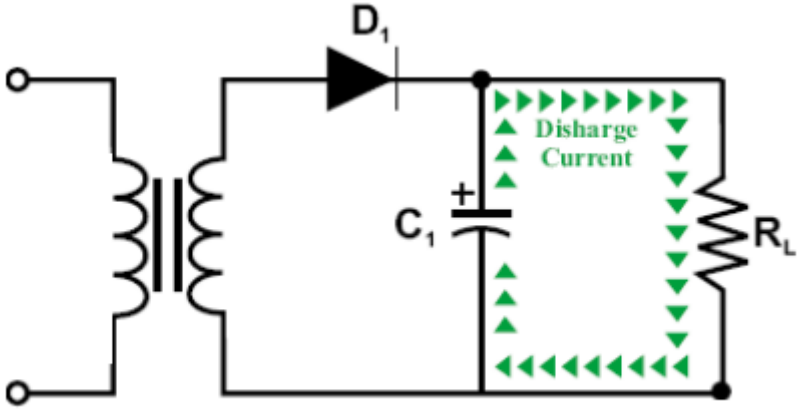
- **During the positive half cycle, diode D1 will conduct, and the capacitor charges rapidly.**



1. Half-Wave Rectifier with Capacitor Filter

Negative Half Cycle

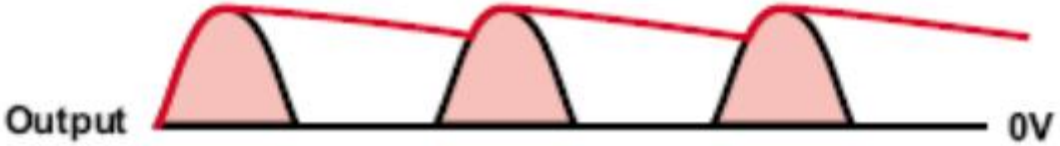
- As the input starts to go negative, D1 turns off, and the capacitor will slowly discharge through the load.



Output Waveforms with Capacitor Filter



(a) Unfiltered output from the half wave rectifier



(b) When the next pulse does arrive, it charges the capacitor back to full charge as shown above. The thick line shows the charge – discharge waveform at the capacitor.

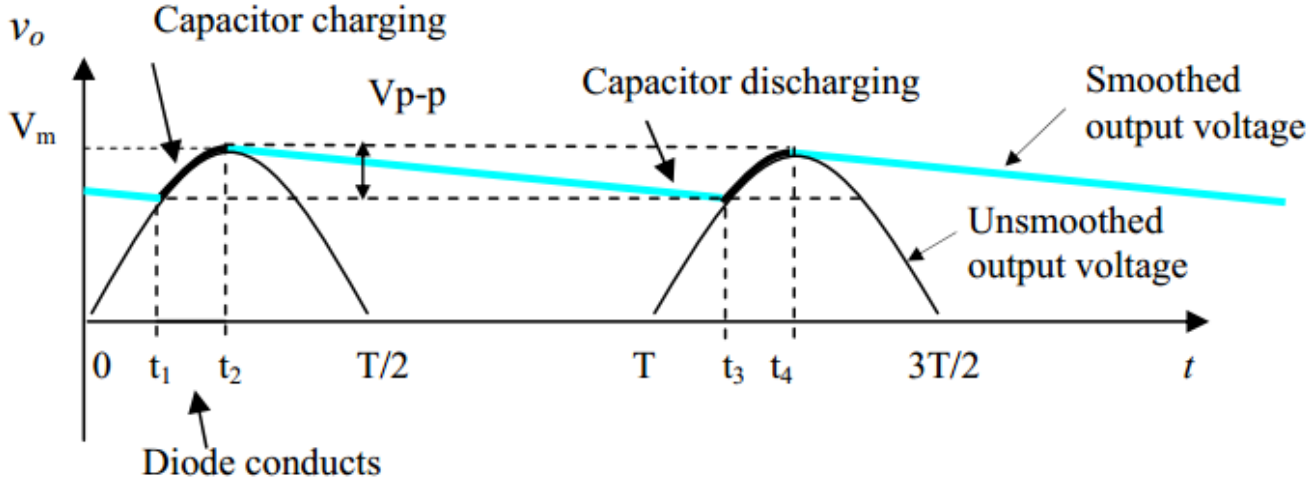


(c) The load sees a reasonably constant DC voltage now, with a ripple voltage on top of it.

Detailed Analysis of Filtering Output

Positive Half Cycle

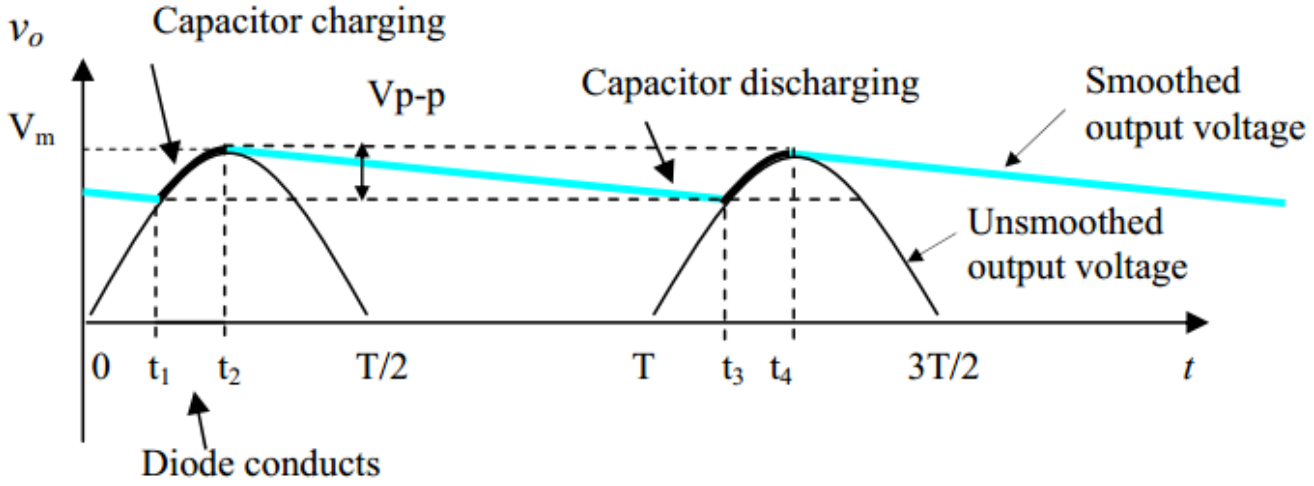
- **During each positive half cycle, the capacitor charges during the interval t_1 to t_2.**
- **During this interval, the diode will be forward biased.**



Detailed Analysis of Filtering Output

Positive Half Cycle

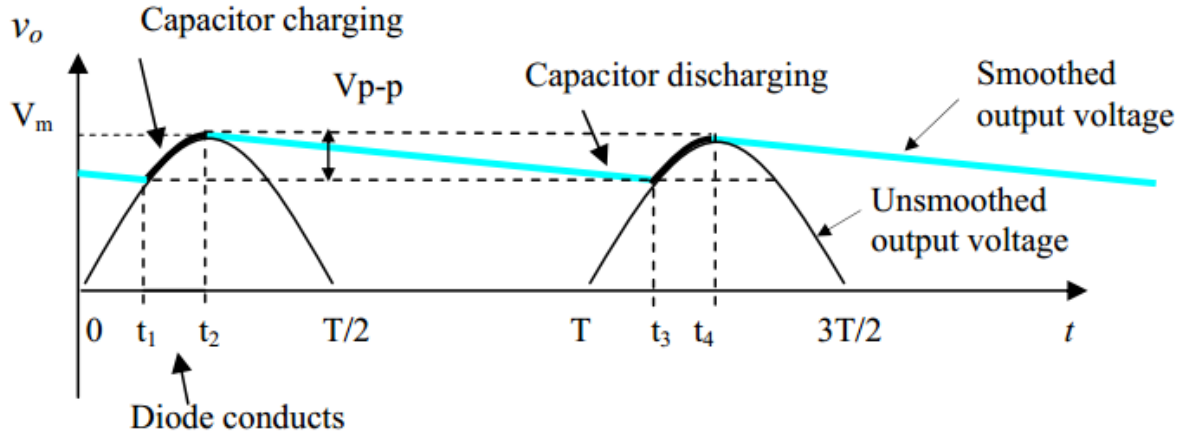
- **Due to this charging, the voltage across the capacitor v_o will be equal to the AC peak voltage V_m on the secondary side of the transformer at t_2 (assuming diode forward voltage drop is zero).**



Detailed Analysis of Filtering Output

Negative Half Cycle

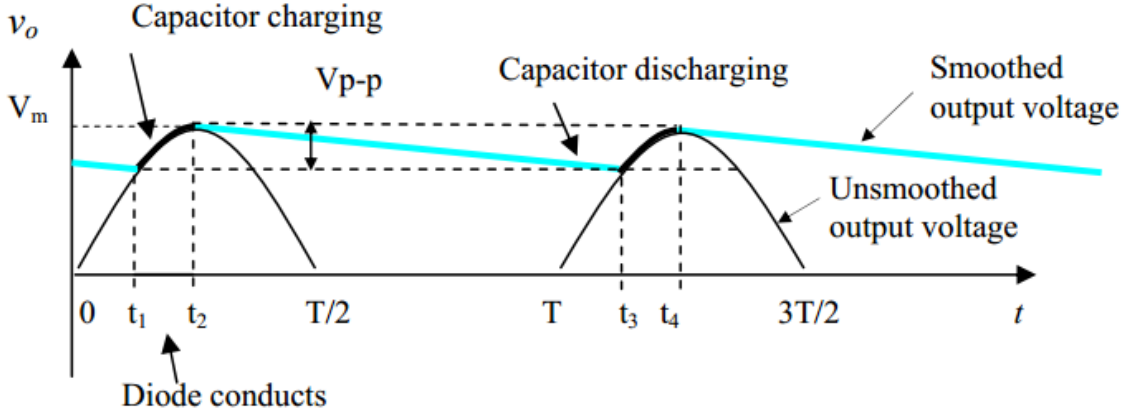
- **The capacitor will supply current to load resistor R_L during time interval t_2 to t_3.**
- **During this interval, diode will be reverse biased since the AC voltage is less than the output voltage v_o.**



Detailed Analysis of Filtering Output

Negative Half Cycle

- **Due to the large energy stored in the capacitor, the capacitor voltage will not reduce much during t_2 to t_3, and the voltage v_o will remain close to the peak value.**
- **As can be seen, addition of the capacitor results in much better quality output voltage.**



Average Load Voltage

- **In practical applications, a very large capacitor is used so that the output voltage is close to the peak value.**
- **The average voltage (also called DC output voltage) across the load can therefore be approximated to:**

$$V_{ave} \cong V_m$$

Calculation of Capacitance Value

- The voltage waveforms show a small AC component called “ripple” present in the output voltage.
- This ripple can be minimized by choosing the largest capacitance value that is practical.
- The capacitor is typically “electrolytic” type, and is very large (several hundreds or even thousands of microfarads).



Electrolytic Capacitor

Calculation of Capacitance Value

- We can calculate the required value of the filter capacitor as follows.
- The charge removed from the capacitor during the discharge cycle (i.e., t_2 to t_3) is:

$$\Delta Q \cong I_L T$$

- Where I_L is the average load current and T is the period of the AC voltage. As the interval t_1 to t_2 is very small, the discharge time can be approximated to T.
- If $V_{p\text{-}p}$ is the peak-to-peak ripple voltage, and C is the capacitance, the charge removed from the capacitor can also be expressed as:

$$\Delta Q = V_{p\text{-}p} C$$

Calculation of Capacitance Value

- From the two above equations, capacitance C can be calculated as:

$$C = \frac{I_L T}{V_{p_p}} \text{ Farad}$$

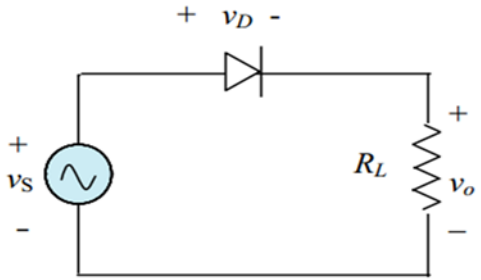
- Our goal is to produce a constant dc output voltage.
- The filter capacitor will remove most of the variations in our rectifier output waveform.

The amount of ripple voltage left by a given filter depends on the three things:

1. Type of rectifier (half or full wave)
2. The capacity of the filter capacitor
3. The load resistance

Example 4

In the circuit of example 1, a 10000μF filter capacitor is added across the load resistor. The voltage across the secondary terminals of the transformer is 230V rms. R_L= **50Ω**



Determine the:

(a) DC output voltage

(b) Load current

(c) The charge supplied by the capacitor to the load

(d) Peak-to-peak ripple in the output voltage

(e) Percentage ripple

(f) The output ripple frequency.

Work Sheet

Solution

(a) DC output voltage

$V_{ave} \cong V_m = 325.3\ V$

(b) The load current

$I_L = \frac{V_{ave}}{R_L} = \frac{325.3}{50} = 6.51\ A$

(c) The charge supplied by the capacitor to the load

This current discharges the capacitor during the interval t_2 to t_3. The time period of the AC voltage = 20ms (for 50 Hz frequency).

Thus, the charge supplied by the capacitor to the load resistance during this interval will be:

$\Delta Q = I_L T = 6.51 \times 20 \times 10^{-3} = 0.1302\ C$

Solution

(d) Peak-to-peak ripple voltage

$$V_{p\text{-}p} = \frac{\Delta Q}{C} = \frac{0.1302}{10000 \times 10^{-6}} = 13.02 \text{ V}$$

(e) Percentage ripple

$$\text{Percentage ripple} = \frac{V_{p_p}}{V_{ave}} \times 100\% = \frac{13.02}{325.3} \times 100\% = 4\%$$

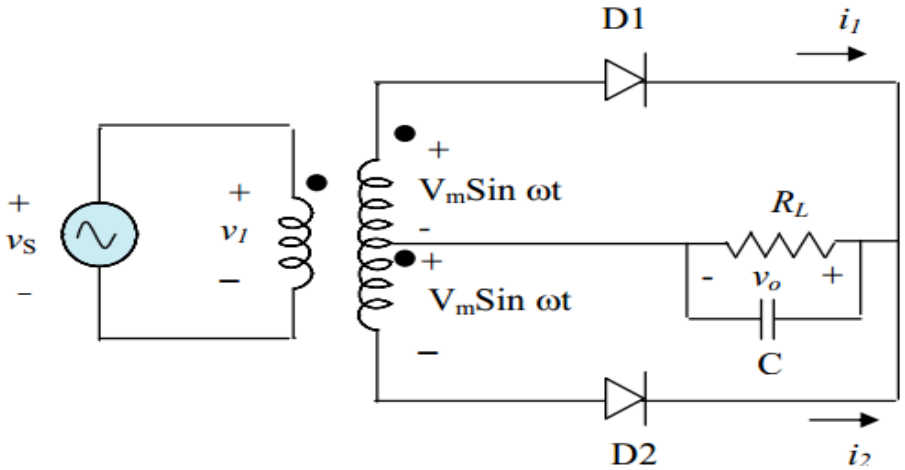
The larger the capacitor value, the smaller the ripple.

Notice that the ripple voltage is now only 4%, compared to 314% when the capacitor is not used.

(f) The ripple frequency is same as before (50 Hz).

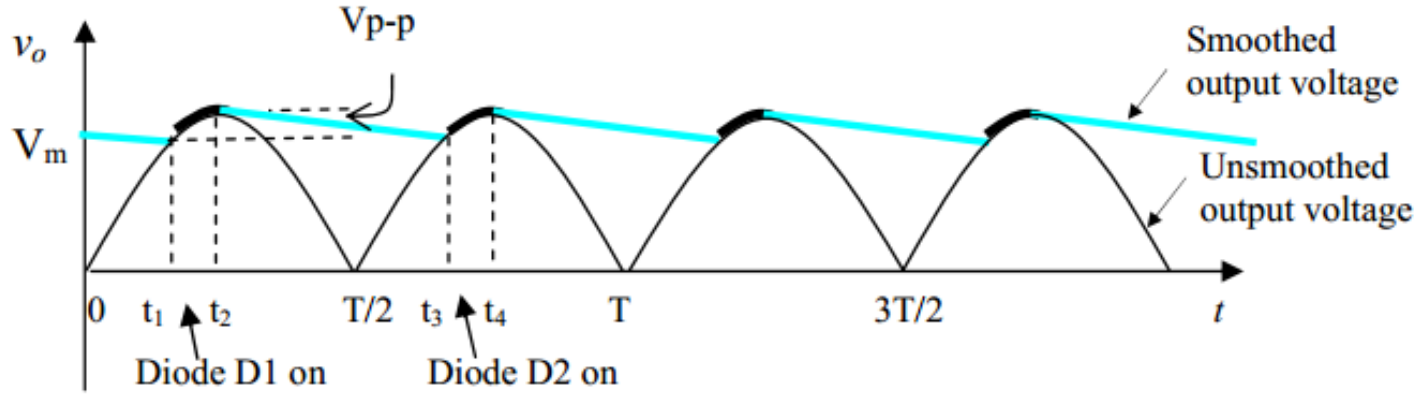
2. Centre-Tapped Rectifier with Capacitor filter

- **Similar to the half-wave rectifier, smoothing is performed by a large value capacitor connected across the load resistance as shown below to act as a reservoir, supplying current to the output when the varying DC voltage from the rectifier is falling.**



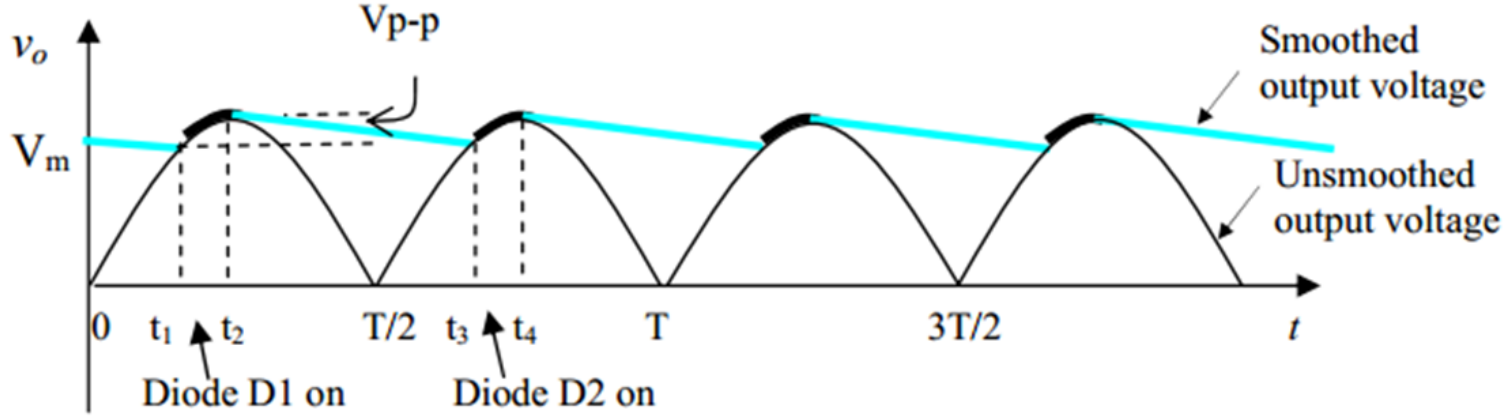
Output Waveform with Capacitor Filter

- **The diagram below shows the unsmoothed varying DC (thin line) and the smoothed DC (thick line).**
- **The capacitor charges quickly near the peak of the varying DC, and then discharges as it supplies current to the output.**



Output Waveform with Capacitor Filter

- **Note that smoothing significantly increases the average DC voltage to almost the peak value.**
- **However, smoothing is not perfect due to the capacitor voltage falling a little as it discharges, giving a small ripple voltage.**



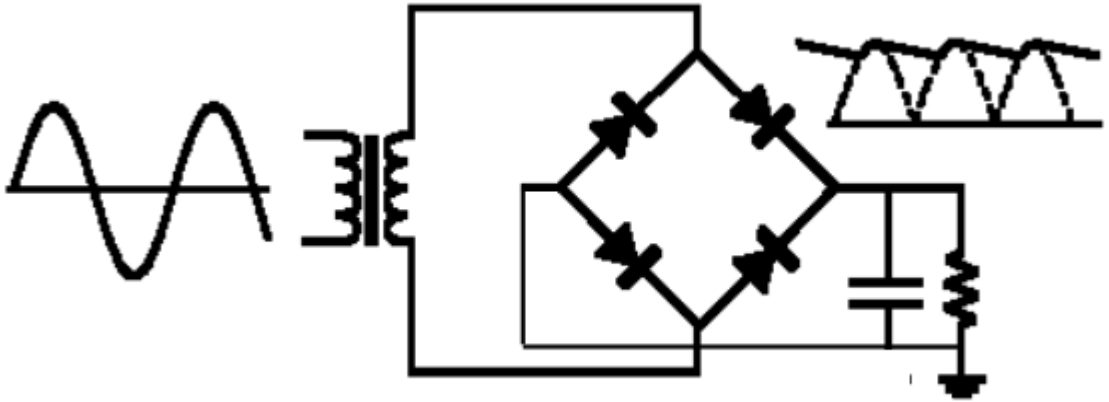
Calculation of Capacitance Value

- For many circuits, a ripple which is 10% of the supply voltage is satisfactory and the equation below gives the required value for the smoothing capacitor.
- In the full-wave circuit, the capacitor discharges for only a half-cycle before being recharged.
- Hence the capacitance required is only half as much in the full-wave circuit as for the half-wave circuit.

$$C = \frac{I_L T}{2V_{p_p}}$$

3. Bridge Rectifier With Capacitor Filter

- **The voltage obtained across the load resistor of the full-wave bridge rectifier described in the previous slides has a large amount of ripple.**
- **A capacitor filter may be added to smoothen the ripple in the output, as shown below.**



2. ZENER DIODE

A Zener diode is a special diode which operates during the reverse biased mode for voltage regulation.



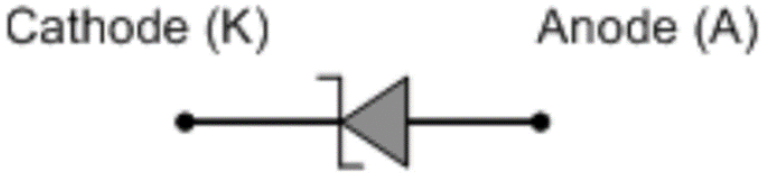
The Zener Diode

- A reverse biased diode passes very little current but will suffer breakdown or damage if the reverse voltage applied across it is made too high resulting in **Zener diodes or breakdown diodes.**
- Zener diodes are basically the same as the standard junction diode but are specially made to have a low pre-determined reverse breakdown voltage, called the **Zener voltage (V_Z).**



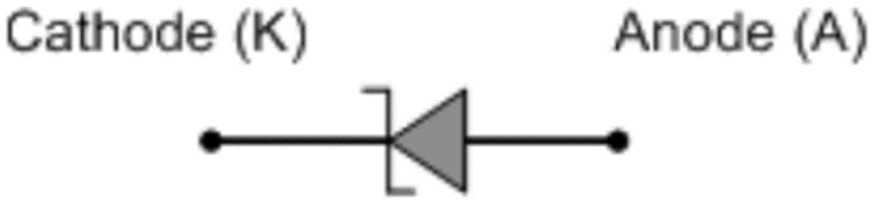
The Zener Diode

- **In the forward direction, it behaves just like a normal signal diode passing current.**
- **When the reverse voltage applied to it exceeds the selected reverse breakdown voltage, a process called avalanche breakdown occurs in the depletion layer.**
- **The current through the diode increases to the maximum circuit value, which is usually limited by a series resistor.**



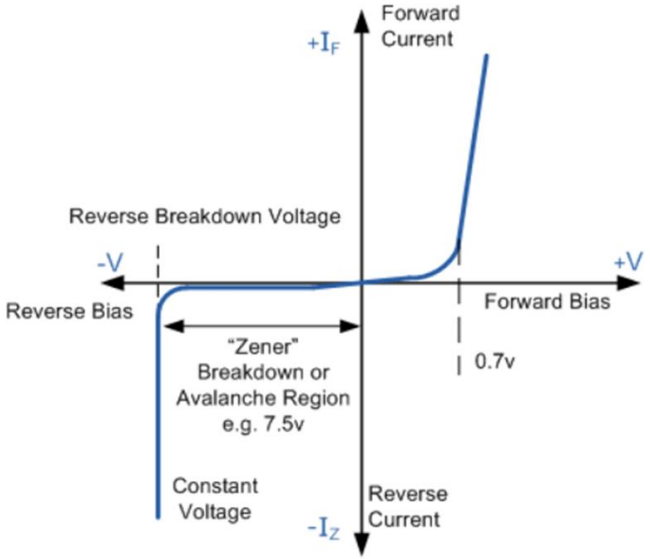
The Zener Diode

- **The point at which current flows can be very accurately controlled (to less than 1% tolerance) in the doping stage of the diode construction giving it a specific Zener.**
- **Breakdown voltage (V_z) ranging from a few volts up to a few hundred volts.**



The Zener Diode

- **Zener Diodes are used in the reverse bias mode, i.e. the anode connects to the negative supply and from its I-V characteristics curve below.**
- **The Zener diode has a region in its reverse bias characteristics of almost a constant voltage regardless of the current flowing through the diode.**

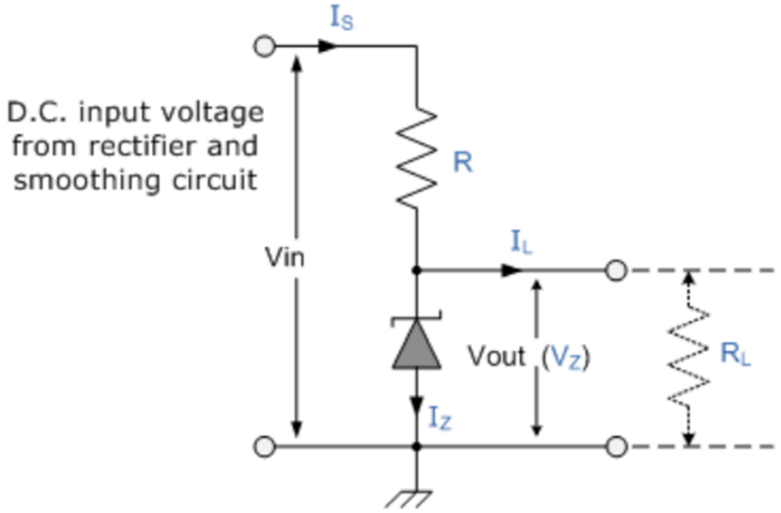


The Zener Diode

- **This voltage across the diode (its Zener voltage,) remains nearly constant even with large changes in current through the diode caused by variations in the supply voltage or load.**
- **This ability to control itself can be used to great effect to regulate or stabilize a voltage source against supply or load variations.**
- **The diode will continue to regulate until the diode current falls below the minimum value in the reverse breakdown region.**

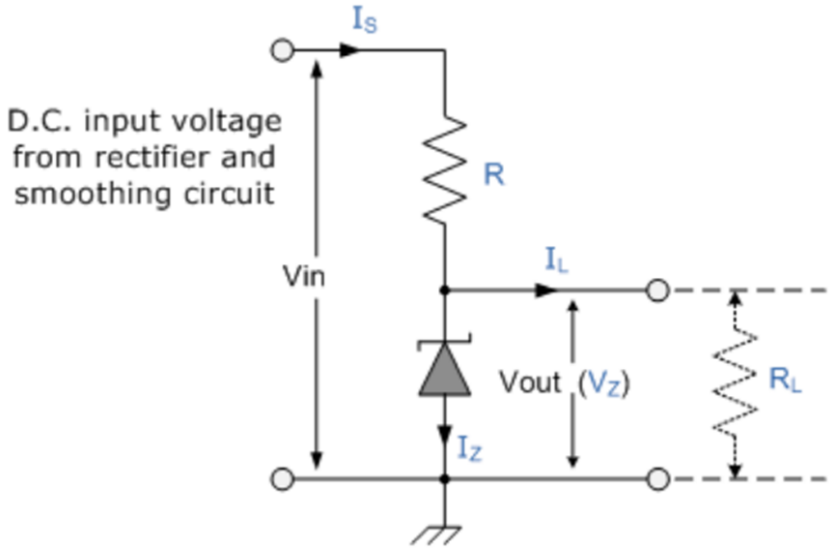
Zener Diode as a Voltage Regulator

- **A voltage regulator is a circuit designed to ensure a constant, stable voltage across a load.**
- **By connecting a simple Zener stabiliser circuit as shown below across the output of the rectifier, a more stable reference voltage can be produced.**



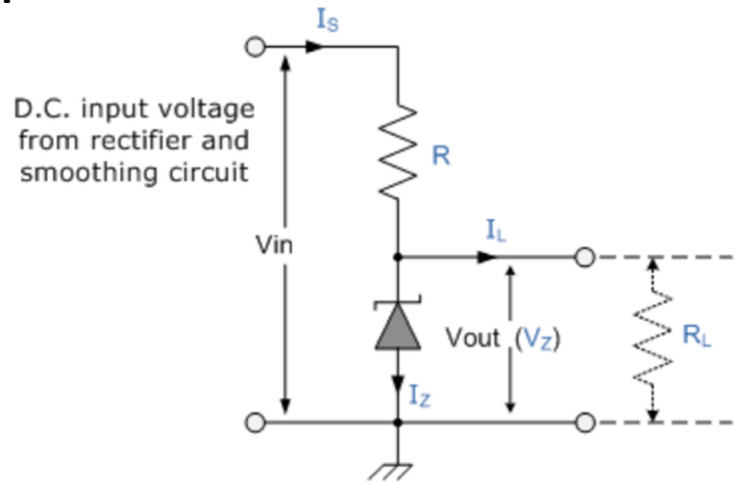
Zener Diode as a Voltage Regulator

- **The resistor is connected in series to limit the current flow to the diode.**
- **The input voltage source, V_{in} is connected across the combination while the stabilised output voltage V_{out} is taken from across the Zener diode.**



Zener Diode as a Voltage Regulator

- **The Zener diode is connected with its cathode terminal connected to the positive rail of the DC supply so it is reverse biased and will be operating in its breakdown condition.**
- **When no load resistance (R_L) is connected to the circuit, no load current (I_L = 0) is drawn and all the circuit current passes through the Zener diode which dissipates its maximum power.**



Zener Diode as a Voltage Regulator

- **Care must be taken when selecting the appropriate value of resistance such that the Zener maximum power rating is not exceeded under this "no-load" condition.**
- **There is a minimum Zener current for which the stabilization of the voltage is effective and the Zener current must stay above this value when operating within its breakdown region at all times.**
- **The upper limit of current is of course dependent upon the power rating of the device.**

Example 1

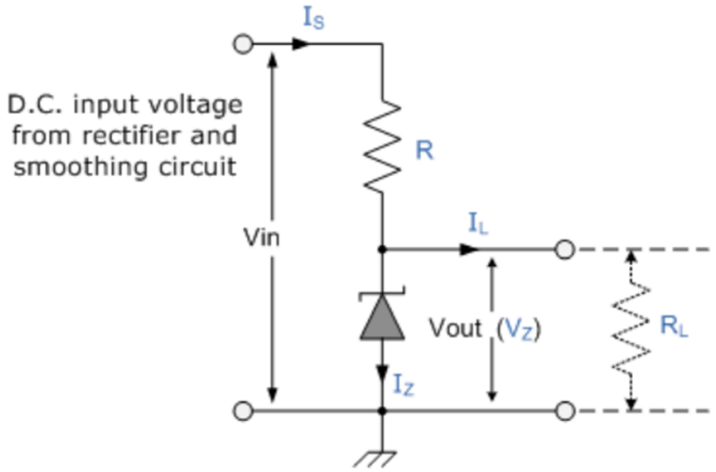
A 5.0V stabilized power supply is required from a 12V DC input. The maximum power rating of the Zener diode is 2W. Using the circuit below calculate:

(a) The maximum current flowing in the Zener diode

(b) The value of the series resistor R.

(c) The load current I_L if a load resistor of 1kΩ is connected across the Zener diode.

(d) The total supply current I_S



Solution

(a) The maximum current flowing in the Zener diode

$$I_{zmax} = \frac{P_z}{V_z} = \frac{2}{5} = 400\text{mA}$$

(b) The value of the series resistor R

$$R = \frac{V_{IN} - V_z}{I_z} = \frac{12 - 5}{400 \times 10^{-3}} = 17.5\ \Omega$$

(c) The load current I_L if a load resistor of 1kΩ is connected across the Zener diode.

$$I_L = \frac{V_z}{R_L} = \frac{5}{1 \times 10^3} = 5\text{mA}$$

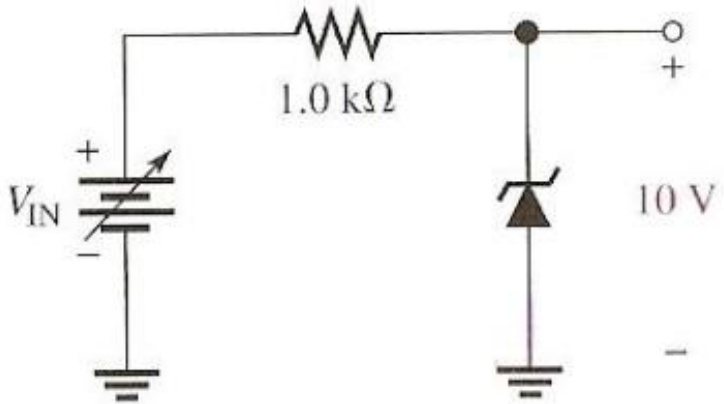
(d) The total supply current I_S

$$I_s = I_z + I_L = 400\text{mA} + 5\text{mA} = 405\text{mA}$$

Example 2

In the circuit below the 10V Zener diode can maintain regulation over a range of current values from 4mA to 40mA.

Calculate the range of input voltage for which the diode can maintain the regulated output.



Solution

The range of input voltage for which the diode can maintain the regulated output.

For the minimum input voltage

$I_z = 4mA$

$V_R = I_z R = 4 \times 10^{-3} \times 1 \times 10^3 = 4V$

$V_{in} = V_R + V_z = 4V + 10V = 14V$

For the maximum input voltage

$I_z = 40mA$

$V_R = I_z R = 40 \times 10^{-3} \times 1 \times 10^3 = 40V$

$V_{in} = V_R + V_z = 40V + 10V = 50V$

The range of input voltage is from 14V – 50V

Percentage Regulation

- The regulation is merit used to specify the performance of a voltage regulator.
- It can be in terms of input (line) or load regulation.

Line regulation

- Line regulation specifies how much change occurs in the output for a given change in the input voltage. It is typically defined as:

$$\text{Line Regulation} = \frac{\Delta V_{out}}{\Delta V_{in}}$$

Percentage Regulation

Load regulation

Load regulation specifies how much change occurs in the output over a certain range of load current values, usually from minimum current (no load) to maximum current (full load). It is normally calculated with the formula:

$$\textbf{Load Regulation} = \frac{V_{\text{no load}} - V_{\text{full load}}}{V_{\text{full load}}} = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

Example 3

Assume a certain 7805 regulator has a measured no load output voltage of 5.18V and a full- load output of 5.15V. What is the load regulation expressed in percentage?

Solution

$$\text{Load Regulation} = \frac{V_{\text{no load}} - V_{\text{full } load}}{V_{\text{full } load}} = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

$$\text{Load Regulation} = \left(\frac{5.18 - 5.15}{5.15}\right) \times 100\% = 0.58\%$$

3. LIGHT EMITTING DIODE

A light-emitting diode is a semiconductor light source that emits light when current flows through it.



Light Emitting Diodes (LEDs)

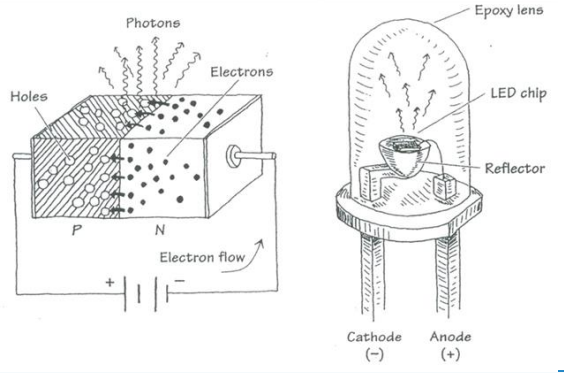
- **LEDs are the most visible type of diode that emits a fairly narrow bandwidth of either visible coloured light, invisible infra-red or laser-type light when a forward current is passed through them.**
- **When the diode is forward biased, electrons from the semiconductor's conduction band combine with holes from the valence band, releasing sufficient energy to produce photons of light.**
- **Because of this thin layer, a reasonable number of these photons can leave the junction and radiate away producing a coloured light output.**

Light Emitting Diodes (LEDs)

- **Normal diodes which are made for signal detection and power rectification are generally made from either Germanium or Silicon semiconductor material.**
- **LEDs are made from compound-type semiconductor materials such as Gallium Arsenide (GaAs), Gallium Phosphide (GaP), Gallium Arsenide Phosphide (GaAsP), Silicon Carbide (SiC) or Gallium Indium Nitride (GaInN).**
- **The exact choice of the semiconductor material used will determine the overall wavelength of the photon light emissions and therefore the resulting colour of the light emitted**

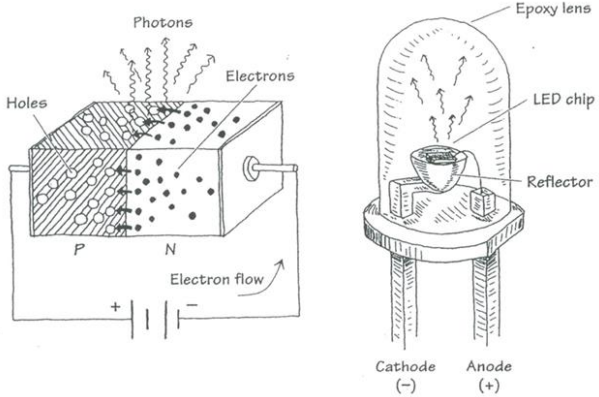
How LEDs work?

- **The light-emitting section of an LED is made by joining n-type and p-type semiconductors together to form a p-n junction.**
- **When the p-n junction is forward-biased, electrons in the n side are excited across the p-n junction and into the p side, where they combine with holes.**
- **As the LED is to be connected in a forward bias condition across a power supply, it should be current limited using a series resistor to protect it from excessive current flow.**



How LEDs work?

- **As the electrons combine with the holes, photons are emitted.**
- **The p-n junction section of an LED is encased in an epoxy shell that is doped with light scattering particles to diffuse light and make the LED appear brighter.**
- **Often a reflector placed beneath the semiconductor is used to direct the light upward.**



How LEDs work?

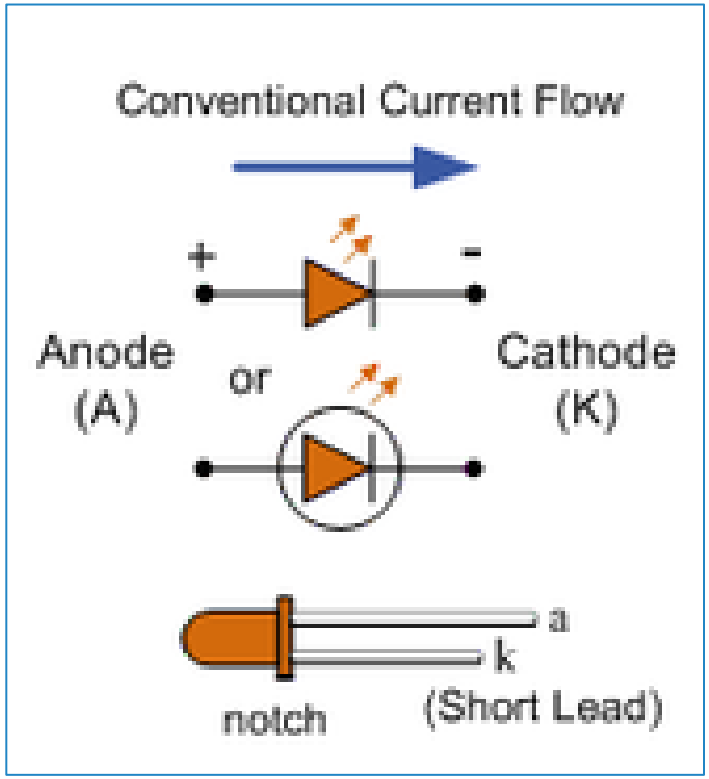
- From the table below we can see that each LED has its own forward voltage drop across the PN-junction.

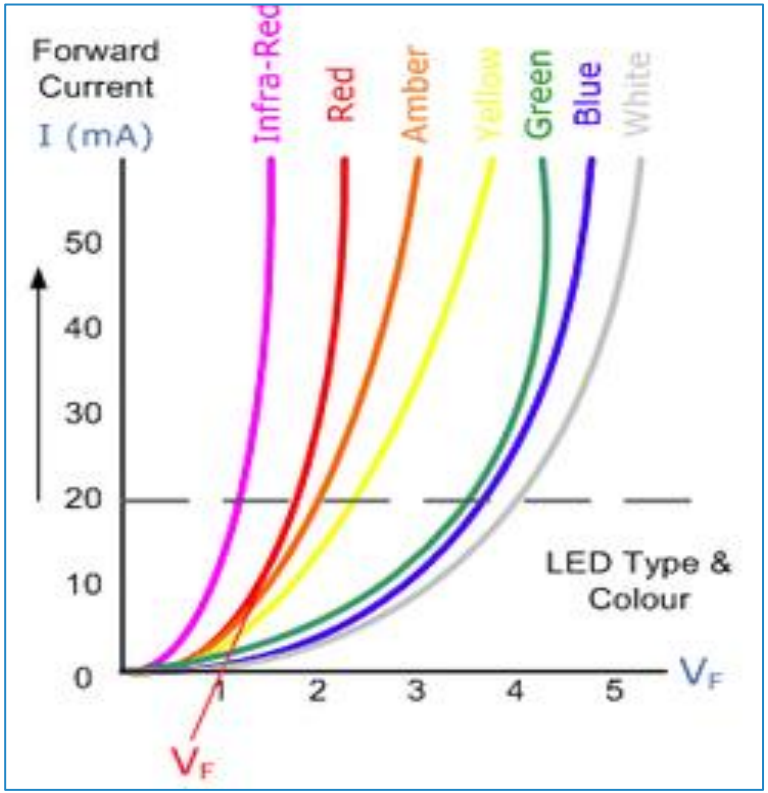
Typical LED Characteristics			
Semiconductor Material	Wavelength	Colour	V_F @ 20mA
GaAs	850-940nm	Infra-Red	1.2v
GaAsP	630-660nm	Red	1.8.v
GaAsP	605-620nm	Amber	2.0v
GaAsP:N	585-595nm	Yellow	2.2v
AlGaP	550-570nm	Green	3.5v
SiC	430-505nm	Blue	3.6v
GaLnN	450nm	White	4.0v

How LEDs work?

- **This parameter, which is determined by the semiconductor material used, is the forward voltage drop for a given amount of forward conduction current, typically for a forward current of 20mA.**
- **In most cases LEDs are operated from a low voltage DC supply, with a series resistor to limit the forward current to a suitable value from say 5mA for a simple LED indicator to 30mA or more where a high brightness light output is needed.**

IV Characteristics of LEDs





Application of LEDs

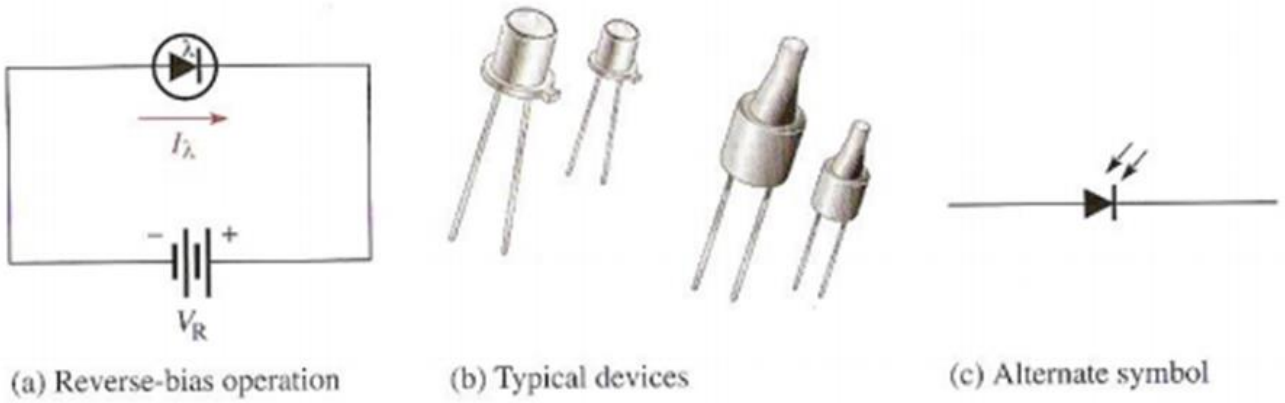
- **LEDs are commonly used for indicator lamps and readout displays on a wide variety of devices.**
- **A common type of display using LEDs is a 7-segment display. A 7-segment LED display provides a very convenient way of displaying information or digital data in the form of Numbers, Letters or even Alpha-numerical characters.**
- **As the name suggests, the means of display consists of 7 individual LEDs (the segments), within one single display package.**

Application of LEDs

- **In order to produce the required numbers or characters from 0 to 9 and A to F respectively, the correct combination of LED segments need to be illuminated on the display.**
- **A standard 7-segment LED display generally has 8 input connections, one for each LED segment and one that acts as a common terminal or connection for all the internal segments.**

Photodiodes

- **The photodiode is a junction device that operates in reverse bias as shown in the figure.**
- **The photodiode has a small transparent window that allows light to strike the p-n junction. Typical photodiodes are shown the figure below**

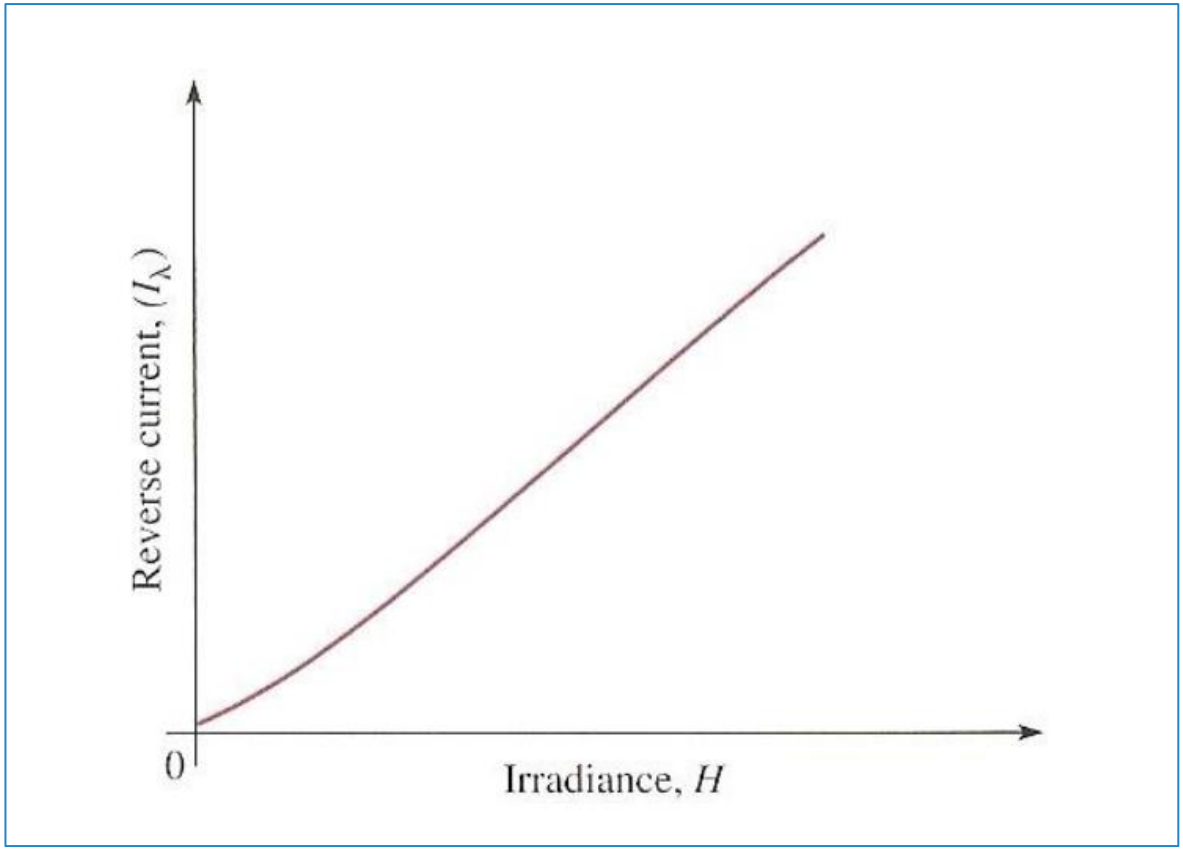


(a) Reverse-bias operation (b) Typical devices (c) Alternate symbol

Photodiodes

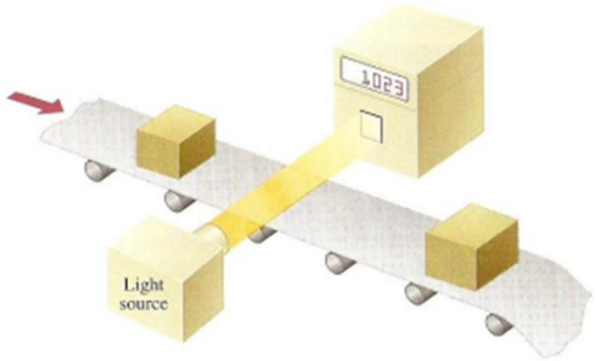
- **In the photodiode, the reverse current increases with the light intensity at the exposed junction.**
- **When there is no incident light, the reverse current, I_λ is almost negligible and is called dark current.**
- **An increase in the amount of light intensity, expressed as irradiance (mW/cm^2), produces an increase in the reverse current as shown by the graph in figure.**

Reverse Current - Irradiance Curve



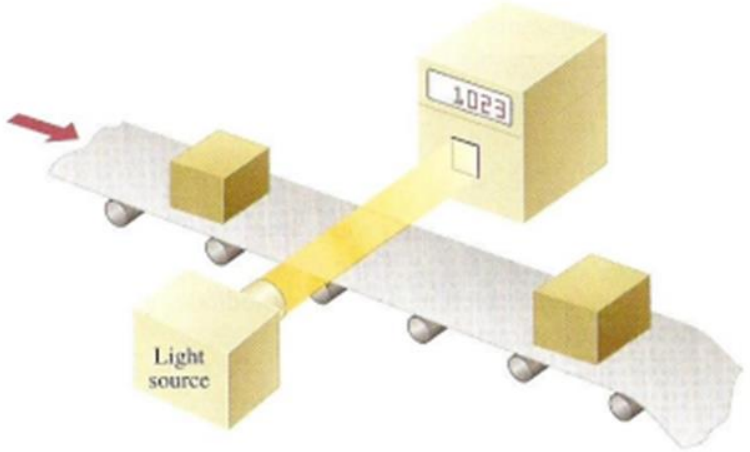
Applications of Photodiodes

- An application of photodiode is depicted in figure below.
- Here, a beam of light continuously passes across a conveyor belt and into a transparent window behind which is a photodiode circuit.
- When the light beam is interrupted by an object passing by on the conveyor belt, the sudden reduction in the diode current activates a control circuit that advances a counter by one.



Applications of Photodiodes

- **The total count of object that has passed that point is displayed by the counter.**
- **This basic concept can be extended and used for production control, shipping, and monitoring of activity on production lines.**



Thanks!

Any questions?